Project Linguistic CAN

Position Paper

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The CAN Bus as a Linguistic System: Reframing Automotive Network Reverse Engineering as Field Linguistics

Abstract

The Controller Area Network (CAN) bus is the dominant intra-vehicle communication protocol, yet the proprietary nature of its signal encodings remains the central obstacle to programmatic vehicle control. Existing approaches to CAN bus reverse engineering treat the problem as one of signal processing or statistical classification (Marchetti and Stabili 1083; Pese et al. 2283). This paper argues that these methods, while effective, mischaracterize the nature of the problem. We propose instead that CAN bus encodings constitute linguistic systems in the structural sense defined by Saussure and formalized by Chomsky, and that the appropriate intellectual framework for reversing them is not signal processing but field linguistics. We present corpus evidence from 86 publicly available DBC files spanning 14 manufacturer families. We demonstrate that manufacturer encoding conventions exhibit the defining properties of natural language families: arbitrary form-meaning mappings, systematic intra-family regularity, inter-family divergence, and morphological compositionality. We propose OBD-II as a Rosetta Stone enabling bilingual correlation between standardized and proprietary encodings, and describe a small language

model architecture that generates command frames for unseen vehicles through internalized linguistic competence rather than explicit lookup.

1. Introduction

Every vehicle manufactured since the early 1990s communicates internally over a Controller Area Network bus. The CAN standard, codified in ISO 11898, specifies the physical and data-link layers of the protocol: differential signaling, priority-based arbitration, cyclic redundancy checking, and a maximum payload of eight bytes per frame. What the standard does not specify is the semantic content of those eight bytes. The interpretation of CAN frame payloads is left entirely to the manufacturer (ISO 11898-1).

These interpretations are documented in proprietary database files known as DBC files, a de facto standard originally developed by Vector Informatik. A DBC file maps CAN arbitration IDs to message names and maps bit positions within each message to named signals with defined scaling factors, offsets, byte orderings, and value ranges. Without the DBC for a given vehicle, raw CAN traffic is semantically opaque. As Verma et al. observe, the prevailing paradigm for passenger vehicle CAN is "security by obscurity," in which OEMs hold signal definitions secret and vary them across years and models (Verma et al. 1).

The dominant approach to resolving this opacity treats it as a signal processing problem. Marchetti and Stabili proposed bit-flip rate analysis for identifying signal boundaries (Marchetti and Stabili 1084). Pese et al. developed LibreCAN, correlating CAN traffic against OBD-II responses and inertial measurements (Pese et al. 2284). Lin et al. introduced ByCAN, integrating byte-level and bit-level clustering to achieve 80% slicing accuracy (Lin et al. 1). Ngo et al. extended this with CANClassify, applying supervised learning to label decoded signals (Ngo et

al. 1). Each advances the state of the art, but each operates within the same implicit framing: CAN encodings are technical signals to be decoded, and signal processing is the appropriate instrument.

This paper proposes a different framing. We argue that manufacturer CAN encodings are not technical signals but linguistic systems, and that the tools of computational linguistics are both more natural and more powerful for the task of reverse engineering them. We develop this argument through structural analysis, corpus evidence, and a concrete architectural proposal. Throughout, we use comparative examples from natural human languages to make the structural parallels legible to readers outside the automotive domain.

2. Structural Parallels Between CAN Encodings and Natural Languages

2.1 Arbitrariness of the Sign

The foundational principle of structural linguistics, identified by Saussure, is the arbitrariness of the sign: the relationship between a linguistic form and its meaning is conventional rather than motivated by any natural correspondence (Saussure 67-69). To make this concrete, consider the sentence "The car stops quickly" expressed in English and in Chinese.

Figure 1. "The car stops quickly" in English

	Subject	Verb	Adverb
English	the car	stops	quickly
Position	1st	2nd	3rd
Grammar	article + noun	conjugated (-s)	suffix (-ly)

Figure 2. Same meaning in Chinese

	Subject	Adverb	Verb	Tense
Chinese	车	很快地	停	了
Translation	car	very-fast [particle]	stop	[completed]
Position	1st	2nd	3rd	4th
Grammar	bare noun	compound + particle	bare stem	aspect particle

Same meaning. No shared vocabulary. Different word order. English conjugates the verb; Chinese uses separate particles. Neither encoding is derivable from the other.

The meaning is identical, but every surface-level property differs: vocabulary, word order, and grammatical machinery. The English word "stops" encodes tense through conjugation (an inflectional suffix); Chinese leaves the verb bare and marks completion with a separate particle. Neither system is more correct. Both are arbitrary conventions adopted by their respective speech communities.

CAN bus encodings exhibit this property exactly. Consider the command "apply maximum brake pressure" as encoded by Toyota and Ford.

Figure 3. "Apply maximum brake pressure" on the CAN bus

	Message ID	Signal name	Bit pos.	Scale	Byte order	Output frame
Toyota	`0x224`	`BrakePressure`	`16`	`0.01`	`little-endian`	`00 00 10 27 00 00 00 00`
Ford	`0x140`	`BRAKE_PRESSURE`	`0`	`0.1`	`big-endian`	`FF FF 00 00 00 00 00 00`
Parallel	*vocabulary*	*vocabulary*	*phonology*	*morphology*	*word order*	*the utterance*

Same meaning: maximum brake pressure. Every column differs. The relationship is structurally identical to Figures 1 and 2: two conventional systems encoding identical semantics through entirely different surface forms.

There is no physical or logical necessity for Ford to encode brake pressure at bit 0 with a scale of 0.1 in big-endian byte order, while Toyota uses bit 16 with a scale of 0.01 in little-endian order. These are arbitrary conventions established by engineering teams and perpetuated by institutional inertia, exactly as linguistic conventions are established by speech communities and perpetuated by shared infrastructure (Saussure 67-69).

2.2 Language Families, Dialects, and Idiolects

Natural languages are organized into genealogical families. Languages within a family share structural features inherited from a common ancestor; languages across families do not. Consider the sentence "the car stops" in Spanish, Portuguese, and Mandarin Chinese.

Figure 4. "The car stops" across three languages

	Article	Subject	Verb
Spanish	El	coche	para
Portuguese	O	carro	para
Mandarin	--	车 (car)	停 (stop)

Spanish and Portuguese share nearly everything: cognate articles, similar nouns, and identical verb forms. Mandarin shares nothing. This is the signature of a language family: high intra-family regularity, high inter-family divergence.

Spanish and Portuguese share cognate articles, similar nouns, and in this case an identical verb form, because both descend from Vulgar Latin and their speakers share geographic and institutional infrastructure (Campbell 190-195). Mandarin shares none of these features because it developed independently. The same structure appears in CAN encodings. Consider "apply maximum brake pressure" across two GM vehicles and one Toyota.

Figure 5. Brake command across two GM vehicles and one Toyota

	Message ID	Signal name	Bit pos.	Scale	Byte order	Output frame
GM Cadillac	`0x0C0`	`BRAKE_PRESSURE`	`0`	`0.1`	`big-endian`	`FF FF 00 00 00 00 00 00`

GM Chevy	0x0C0	BRAKE_PRESSURE	0	0.1	big-endian	FF FF 00 00 00 00 00 00
Toyota	0x224	BrakePressure	16	0.01	little-endian	00 00 10 27 00 00 00 00

The two GM vehicles are identical in every column: same message ID, same signal name, same bit position, same scale, same byte order, same output. Toyota differs in every column. This mirrors Figure 4: Spanish and Portuguese are mutually intelligible; Mandarin is a different language entirely.

The economic forces producing this homogeneity within GM are the same forces producing dialectal convergence in human communities: shared infrastructure. GM reuses electronic control units across its product lines because the tooling investment is prohibitive. Diagnostic tools must operate across the portfolio. Supplier relationships enforce component standardization. These function as the CAN bus equivalents of roads, schools, and broadcast media: infrastructure that enforces communicative convergence (Labov 13-15).

2.3 Morphological Structure

In natural language, morphology describes how minimal meaningful units (morphemes) combine to form words. Consider the English word "unhappiness."

Figure 6. Morphological decomposition of "unhappiness"

Morpheme 1: prefix	Morpheme 2: root	Morpheme 3: suffix
un-	happi-	-ness
negation	*core meaning*	*converts adjective to noun*
Position: before root	Position: center	Position: after root

Each morpheme occupies a defined position, carries a defined function, and transforms the composite meaning according to productive rules (Haspelmath and Sims 2).

CAN frames exhibit analogous compositional structure. A CAN frame's eight-byte data payload is composed of multiple signals, each occupying a defined bit range with a defined encoding. Consider a single GM powertrain message (ID 0x0C0) containing three packed signals.

Figure 7. Signal decomposition of CAN message 0x0C0

Signal 1: bits 0-15	Signal 2: bits 16-23	Signal 3: bits 24-31
`BRAKE_PRESSURE`	`BRAKE_PEDAL_POS`	`ABS_ACTIVE`
physical pressure (kPa)	*pedal position (%)*	*ABS engaged (0/1)*
Scale: 0.1, 16-bit	Scale: 0.39, 8-bit	Scale: 1, 1-bit

Each signal occupies a defined bit range within the payload, carries a defined function, and uses a scaling rule to transform raw bits into physical values. The structure is compositional in exactly the way morphological structure is compositional: position, function, and transformation rule.

The arbitration ID functions as a lexical category marker, grouping related signals into a coherent semantic unit. Within the payload, each signal is a morpheme: its bit position parallels phonological placement, its scale factor and offset perform the role of morphophonological rules that transform an underlying representation into a surface form, and its byte order parallels canonical word order. This compositionality is regular within manufacturer families and arbitrary across them, exactly as morphological rules are regular within a language and arbitrary across unrelated languages.

2.4 Pragmatics and Context-Dependence

Natural language utterances derive substantial meaning from pragmatic context. The same surface form can carry different meanings depending on the situation of its utterance (Levinson 97-100). Consider the sentence "Can you pass the salt?"

Figure 8. Context-dependent interpretation of "Can you pass the salt?"

Context	Surface form	Meaning
At dinner	Can you pass the salt?	Please hand me the salt
Medical exam	Can you pass the salt?	Are you physically able to do this?

Identical utterance, different meaning. The context determines the interpretation, not the words themselves.

CAN frames exhibit comparable context-dependence. A frame on a given arbitration ID may encode one set of signals during normal operation and a completely different set when the vehicle is in diagnostic mode.

Figure 9. Context-dependent interpretation of CAN message 0x7DF

Vehicle state	Frame on ID 0x7DF	Interpretation
Normal driving	0x7DF: 02 01 0D 00 00 00 00 00	Not present on bus
OBD-II session	0x7DF: 02 01 0D 00 00 00 00 00	Request vehicle speed (PID 0x0D)

The diagnostic arbitration ID 0x7DF is only meaningful when an OBD-II session is active. Multiplexed messages extend this further: a multiplexer signal within the payload selects which interpretation applies to the remaining bytes.

Understanding this pragmatic layer requires the same kind of contextual reasoning that natural language interpretation demands. A CAN reverse engineering system must know not just what a

signal means in isolation, but when that meaning applies. This is a fundamentally linguistic competence, not a signal processing competence.

3. OBD-II as a Rosetta Stone

The Rosetta Stone enabled the decipherment of Egyptian hieroglyphics because it presented the same text in three scripts, one of which (Greek) was already understood. The On-Board Diagnostics II (OBD-II) standard, mandated in all US vehicles since 1996 under SAE J1979, provides an analogous bridge. OBD-II defines standardized Parameter IDs (PIDs) for engine speed, vehicle speed, throttle position, and dozens of other parameters, encoded uniformly regardless of manufacturer. These queries and responses travel over the same physical CAN bus that carries proprietary messages.

By simultaneously sniffing raw CAN traffic and issuing OBD-II queries, one can correlate standardized responses against proprietary encodings. Consider a query for vehicle speed.

Figure 10. OBD-II as bilingual informant: vehicle speed at 60 km/h

Source	Arbitration ID	Encoding	Decoded value
OBD-II response	`0x7E8`	`PID 0x0D, byte 3`	60 km/h
Ford proprietary	`0x217`	`Bits 0-15, scale 0.01`	60 km/h
Toyota proprietary	`0x1B0`	`Bits 8-23, scale 0.03`	60 km/h

The OBD-II response provides the known value. The proprietary messages, captured simultaneously, encode the same value using manufacturer-specific conventions. The bilingual correlation identifies the signals without physical measurement.

This method is structurally identical to field linguistics methodology. A linguist studying an undocumented language works with a bilingual informant: a speaker who knows both the target language and a language the researcher already speaks. OBD-II functions as this bilingual

informant. It speaks both the universal diagnostic language and the manufacturer's proprietary language. Pese et al. operationalized exactly this principle in LibreCAN, though without framing it in linguistic terms (Pese et al. 2285-2287).

The limitations of OBD-II as an informant are also linguistically predictable. OBD-II covers only parameters relevant to emissions compliance. It does not expose steering angle, individual wheel speeds, or transmission state. This is analogous to a bilingual informant fluent in everyday vocabulary but ignorant of technical terminology. The informant provides an entry point, not a complete dictionary. Extending coverage requires the same techniques linguists use after initial contact: pattern extension, analogical reasoning, and corpus analysis.

4. Corpus Evidence from the OpenDBC Repository

4.1 Description of the Corpus

To evaluate the linguistic framing empirically, we analyzed the commaai/opendbc repository. The corpus contains 116 DBC files, of which 86 parsed successfully. The 86 files span 14 manufacturer families: Ford (6 files, 5,250 signals), General Motors (9 files, 3,617 signals), Toyota (10 files, 702 signals), Honda (27 files, 313 signals), Hyundai/Kia (4 files, 2,496 signals), Tesla (4 files, 1,095 signals), Subaru (7 files, 425 signals), Nissan (3 files, 178 signals), Stellantis (4 files, 239 signals), BMW (1 file, 165 signals), Mercedes-Benz (1 file, 97 signals), Mazda (2 files, 35 signals), Volkswagen Group (1 file, 1,442 signals), and Volvo (1 file, 147 signals).

4.2 Signal Classification and Control Coverage

We classified every signal by functional category using keyword matching against signal names, mapping to six categories: brake, throttle, steering, speed, transmission, and lighting. Of the 86 parsed files, 65 contained at least one controllable signal, and 28 contained all three (brake, throttle, and steering), representing sufficient coverage for full directional vehicle control. From these files, we generated 3,408 structured training pairs mapping vehicle metadata and observed CAN traffic to command frames.

4.3 Intra-Family Regularity

The most consequential finding is the degree of encoding regularity within manufacturer families. To illustrate, compare the brake encoding across three GM vehicles on the Global A platform.

Figure 11. Brake encoding across three GM Global A platform vehicles

	Message ID	Signal name	Bit pos.	Scale	Byte order
GM Cadillac CT6	`0x0C0`	`BRAKE_PRESSURE`	`0`	`0.1`	`big-endian`
GM Chevy Bolt	`0x0C0`	`BRAKE_PRESSURE`	`0`	`0.1`	`big-endian`
GM GMC Sierra	`0x0C0`	`BRAKE_PRESSURE`	`0`	`0.1`	`big-endian`

Every column is identical. The three vehicles speak the same dialect with no variation. This is the CAN equivalent of American English, British English, and Australian English: mutually intelligible, near-identical, produced by shared institutional infrastructure.

These are not regularities imposed by physics. No physical law requires brake pressure to occupy bit 0 or to use a scale of 0.1. These are stylistic conventions, analogous to a speech community's preferences for word order and affixation patterns. They are regular because the same engineering teams, toolchains, and supplier relationships produce them.

4.4 Inter-Family Divergence

While intra-family regularity is high, inter-family divergence is equally pronounced. As demonstrated in Figure 3, Ford and Toyota encode brake pressure using different arbitration IDs, bit positions, scale factors, byte orders, and signal names. The functional semantics are identical; the surface forms share nothing. This pattern of intra-family regularity and inter-family

divergence is the diagnostic signature of a language family structure, confirmed consistently across all 14 families in the corpus.

5. A Small Language Model for CAN Frame Generation

5.1 Theoretical Motivation

If CAN encodings are linguistic systems in the structural sense established above, then the appropriate computational architecture for generating CAN commands is not a classifier or a lookup table but a generative language model: a system that has internalized the productive grammar of the target language and can produce novel, contextually appropriate utterances.

This is the computational analog of the distinction Chomsky draws between competence and performance: competence is the internalized knowledge of a language that enables a speaker to produce and comprehend an unbounded number of novel sentences (Chomsky 4). A speaker of English can understand a sentence never previously encountered because the speaker has internalized the generative rules of English. A language model trained on CAN encodings can produce a command frame for a vehicle it has never encountered, provided the encoding conventions of that vehicle are sufficiently similar to those of vehicles in the training corpus. The intra-family regularity documented in Section 4 indicates this condition is generally satisfied.

The precedent for applying language models to non-natural-language sequences is established. Madani et al. trained ProGen on 280 million protein sequences and demonstrated that it could generate novel, functional proteins with catalytic efficiencies comparable to natural enzymes, including sequences with as low as 31.4% identity to any known natural protein (Madani et al. 1099). Rives et al. showed that unsupervised language modeling at evolutionary

scale captures structural and functional properties of proteins (Rives et al. 1). CAN encodings exhibit the same structural properties that make these applications effective: arbitrary but systematic form-function relationships, compositional structure, and distributional regularities that generalize across the corpus.

5.2 Architecture and Training

We propose fine-tuning a pre-trained small language model of 1.5 billion parameters using quantized low-rank adaptation (QLoRA) on the CAN encoding corpus described in Section 4. The base model, Qwen2.5-1.5B-Instruct, was selected for strong performance on structured output tasks, native handling of hexadecimal token sequences, and compatibility with the Hailo-10H edge inference accelerator, enabling deployment at approximately 10 tokens per second at 2.5 watts. QLoRA reduces trainable parameters from 1.5 billion to approximately 20 million (0.6% of the total) by freezing base weights at 4-bit precision and training only low-rank adapter matrices at key attention layers (Dettmers et al. 10088).

Each training example consists of three components: vehicle metadata (manufacturer, model, platform), a sample of observed CAN traffic representing ambient bus activity, and an operator intent. The target output is the CAN frame or set of frames that executes the specified intent. The model learns to generate the correct output conditioned on the vehicle's "accent": the ambient traffic that reveals its encoding conventions. This architecture is structurally a translation system. The ambient traffic is the source text. The intent is the translation instruction. The output frame is the translation.

5.3 Evaluation Methodology

The evaluation methodology follows directly from the linguistic framing. We hold out entire manufacturer families from training. The model is trained on encodings from Ford, GM, Toyota, Honda, Tesla, and nine other families, and evaluated on Subaru and Mazda. If the model achieves non-trivial accuracy on the held-out families, it demonstrates generalized encoding competence (linguistic competence) rather than memorized mappings (rote vocabulary). The primary metric is exact byte match for safety-critical commands. Partial credit is measured as byte-level overlap but does not constitute success for safety-critical intents.

6. Relationship to Prior Work

6.1 CAN Bus Reverse Engineering

Marchetti and Stabili proposed bit-flip rate analysis for extracting signal boundaries (1084-1086). Pese et al. developed LibreCAN, combining OBD-II correlation, inertial measurement matching, and body event reduction (2285-2290). Lin et al. achieved 80.21% slicing accuracy through byte-level clustering (2). Ezeobi et al. applied unsupervised machine learning to CAN classification (50-56). Ngo et al. extended this with supervised labeling (1-4). Each treats CAN encoding as a technical artifact to be decoded through measurement. None frames the problem as linguistic or proposes that the regularities their algorithms exploit are linguistic in character.

6.2 Language Models for Non-Natural-Language Domains

Madani et al. trained ProGen on 280 million protein sequences and generated novel functional proteins (1099-1106). Rives et al. showed unsupervised pre-training on 250 million sequences captures secondary structure and biological function (1). In code generation, language models routinely produce syntactically and semantically correct programs. The common thread is that each domain's sequences exhibit the structural properties that make language modeling effective. CAN encodings satisfy all of these criteria, yet to our knowledge no prior work has proposed CAN reverse engineering as a language modeling problem.

6.3 The ELINT Precedent

Electronic intelligence analysis classifies unknown radar systems by decomposing emissions into standardized parameters and comparing signatures against a database. The critical difference: in ELINT, the physics is the signal. A radar's pulse characteristics are directly determined by physical design constraints. In CAN encoding, an arbitrary human-designed layer sits between the physics and the signal. Brake pressure is a physical quantity; its CAN encoding is a human convention. This additional indirection is what makes the CAN problem linguistic rather than physical.

7. Implications and Future Directions

If the linguistic framing is correct, several consequences follow. First, the primary bottleneck in CAN reverse engineering is corpus size, not computation. Every DBC file recovered expands the model's vocabulary across the entire manufacturer family. This creates a network effect: each data point benefits all vehicles in the same family.

Second, accuracy should correlate with family coverage in the training corpus rather than total training examples. A thousand examples from one Ford model should improve performance on all Fords more than a thousand examples across ten unrelated manufacturers. This prediction is testable.

Third, cross-lingual transfer learning techniques from NLP may apply. Models trained extensively on well-documented families might achieve non-trivial performance on poorly documented families by exploiting structural universals in CAN encoding design.

Fourth, any protocol in which encoding is determined by human convention rather than physical necessity is, in a structural sense, a language. This insight may extend beyond automotive CAN to industrial SCADA systems, avionics buses, building automation networks, and military communication standards.

8. Conclusion

This paper has argued that the CAN bus is better understood as a linguistic system than as a technical encoding amenable to signal processing. We have shown that manufacturer CAN encodings exhibit the defining structural properties of natural languages: arbitrary form-meaning mappings, genealogical family structure, morphological compositionality, and pragmatic context-dependence. We have demonstrated these properties through both formal argument and concrete comparative examples drawn from human languages and CAN encoding data.

The practical significance of this reframing is that signal processing approaches to CAN reverse engineering are per-vehicle and reactive, while the linguistic approach is per-family and generative. The model does not need to encounter a specific vehicle to command it; it needs to have learned the language that vehicle speaks. This is the difference between carrying a phrasebook for every city and learning the language well enough to speak it anywhere.

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